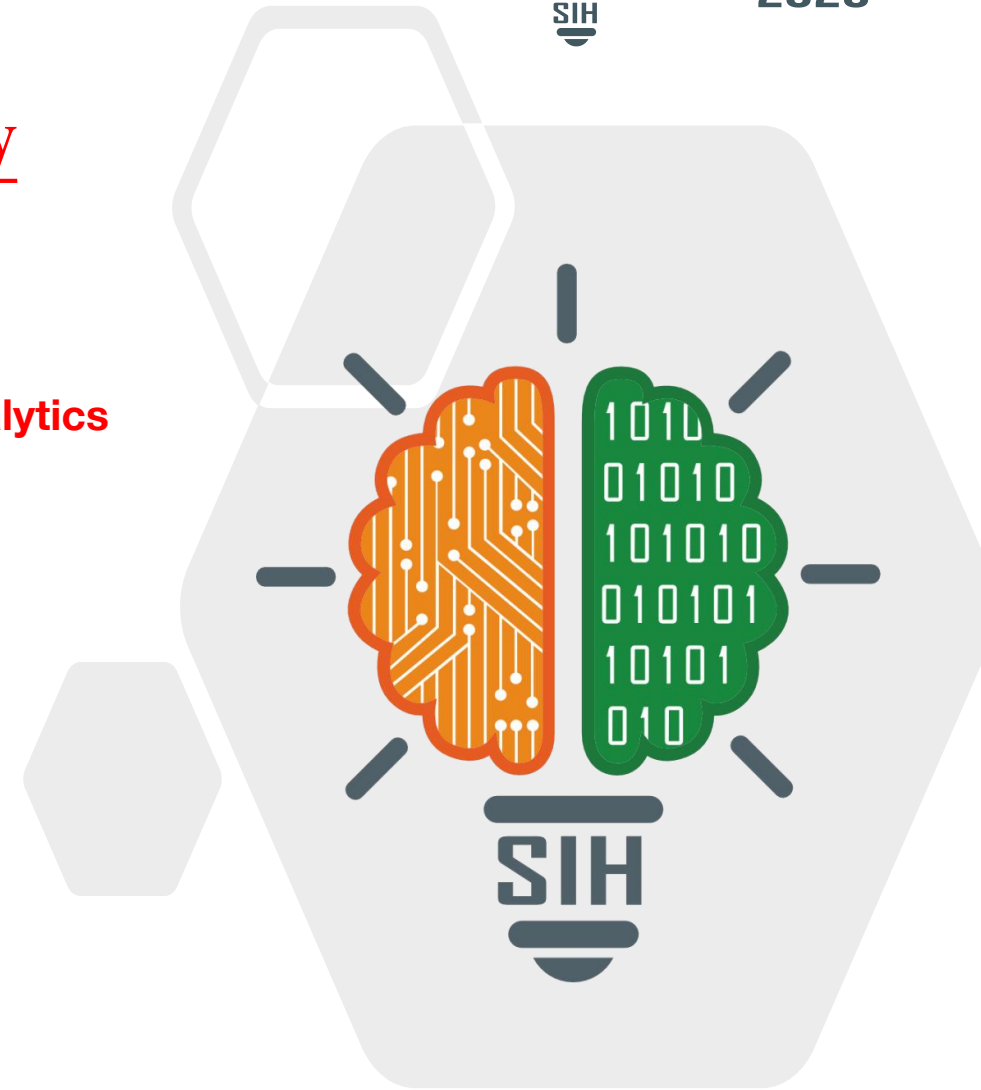


Team - Circuit Crew

- Problem Statement ID – **26187**
- Problem Statement Title - **AI-Based Intelligent Video Analytics**
Platform for Border Surveillance using existing CCTV infrastructure.
- Theme - **Blockchain & Cybersecurity**
- PS Category - **Software**
- Team ID - **KRMU136**
- Team Name : **Circuit Crew**

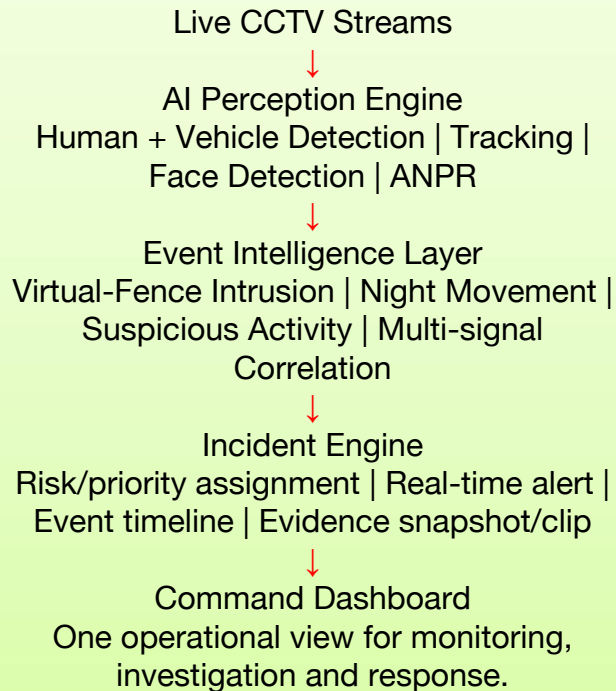


AI- Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV infrastructure.

Core Idea -

IBVAP converts existing IP-based CCTV cameras into an AI-powered border surveillance network without replacing the deployed camera infrastructure.

How the Solution Works



How It Addresses the Problem

From passive video to actionable intelligence

- Removes the need for continuous manual monitoring of every camera feed.
- Reuses existing CCTV infrastructure instead of requiring smart-camera replacement.
- Detects and tracks relevant objects automatically in real time.
- Converts raw video into structured security events with location, time, object and evidence.
- Enables faster identification and prioritisation of potential border incidents.
- Designed as a scalable software layer that can integrate with existing command-and-control systems.

What Makes It Different

Not another CCTV dashboard — an event-driven surveillance system.

Camera-Agnostic

Works as an intelligence layer over existing IP cameras.

Context-Aware Detection

A person alone is not automatically an incident; the system combines object + location + movement + time + behaviour to identify meaningful events.

Incident-Centric Workflow

Instead of asking operators to watch video continuously, the platform brings high-priority events to the operator.

Evidence-First

Every critical alert can be linked with its timestamp, camera, detection metadata and captured evidence for investigation and audit.

Cybersecurity by Design

Access control, audit trails and tamper-evident event/evidence records protect the integrity of surveillance data.

TECHNOLOGY STACK

AI / Computer Vision

Python • OpenCV • YOLO

Real-time person and vehicle detection from IP-camera streams

ByteTrack / DeepSORT

Persistent multi-object tracking across video frames

OCR Engine

Number-plate detection + text recognition for ANPR

Face Recognition Module

Face detection and identity matching where authorized

Video & Backend

RTSP / ONVIF • GStreamer

Ingest and manage existing IP-CCTV streams

FastAPI • Python

Low-latency backend APIs and AI-service integration

WebSockets

Real-time alerts and live event updates

Data & Dashboard

PostgreSQL

Camera, event, incident and audit metadata

Redis

Fast event/alert handling and temporary state

React.js

Command & Control dashboard

Docker

Portable deployment across edge/server environments

Security & Deployment

RBAC + Encryption + Audit Logs

Controlled access to sensitive surveillance data

SHA-256 / Tamper-Evident Evidence Hashing

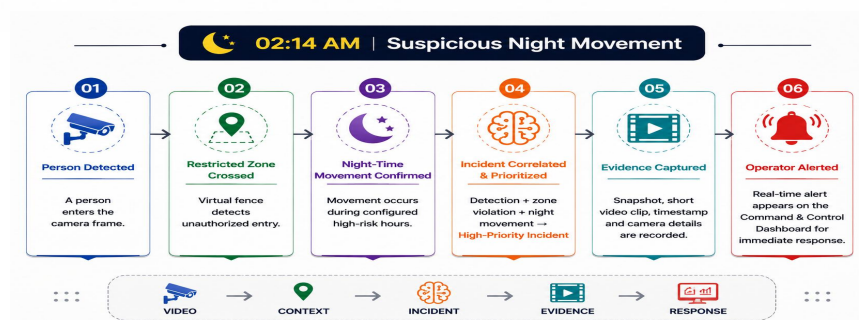
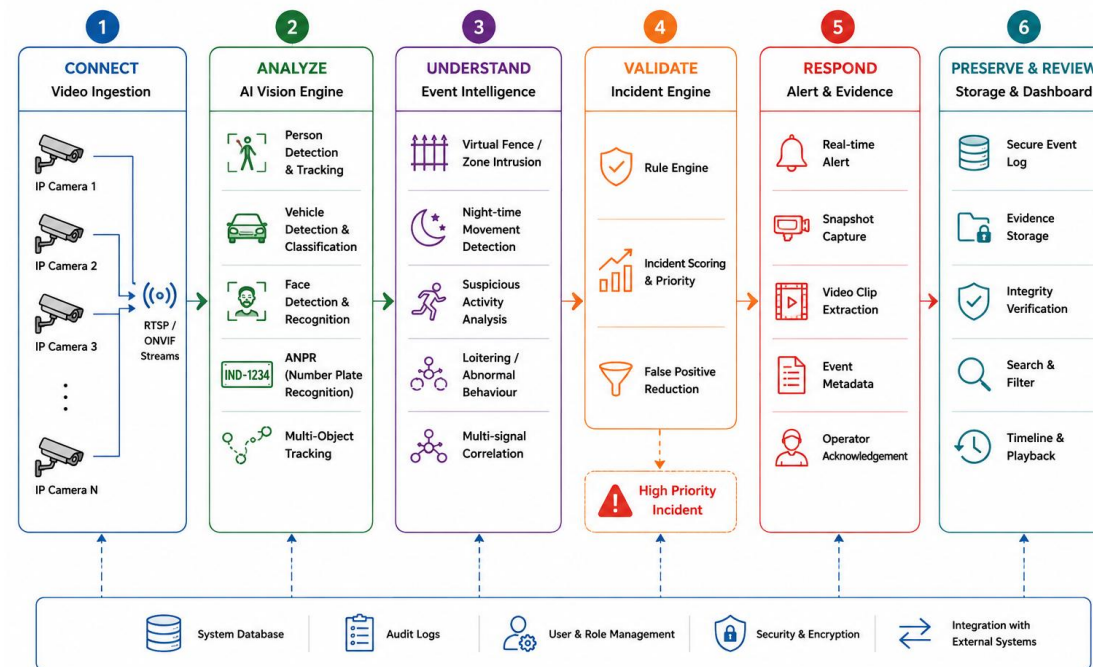
Integrity verification for captured evidence

Edge GPU + Central Server

Local inference for low latency; centralized monitoring and management



METHODOLOGY / IMPLEMENTATION FLOW



Why IBVAP is Practical to Deploy

1. FEASIBILITY

- **Existing CCTV Infrastructure**
 - → Uses standard IP-based CCTV streams
 - → No complete camera replacement required
- **Software-Defined Intelligence**
 - → AI capabilities added through software
 - → Reduces dependency on dedicated smart-camera hardware
- **Edge + Central Architecture**
 - → Critical processing can happen near the camera
 - → Central system handles monitoring, incidents and analytics
- **Modular Implementation**
 - → Person detection, vehicle analytics, ANPR and tracking can be deployed independently
 - → New analytics can be added without redesigning the complete platform
- **Scalable Deployment**
 - → Prototype can start with a small camera group
 - → Architecture can expand to multiple BOPs and command centers

2. CHALLENGES & RISKS

- **Low-Light Conditions**
 - → Night surveillance may reduce detection accuracy
- **Network Reliability**
 - → Remote border locations may have intermittent connectivity
- **Computational Load**
 - → Multiple live streams require significant GPU/CPU resources
- **False Positives**
 - → Incorrect alerts may cause operator fatigue
- **Camera Diversity**
 - → Different resolutions, angles and environments affect model performance
- **Sensitive Data**
 - → Video, faces and vehicle information require controlled access and protection

3. MITIGATION STRATEGY

- **Low-Light → Adaptive Processing**
 - → Image enhancement + night-specific detection + confidence thresholds
- **Network → Edge Intelligence**
 - → Process locally and transmit important events/evidence when connectivity is limited
- **Compute → Optimized AI Pipeline**
 - → GPU acceleration + frame sampling + parallel processing
- **False Alerts → Multi-Signal Validation**
 - → Object + zone + time + movement → stronger incident confidence
- **Camera Variation → Site Calibration**
 - → Camera-specific zones, thresholds and configuration
- **Security → Secure Evidence Pipeline**
 - → Encryption + RBAC + audit logs + evidence integrity verification

TARGET AUDIENCE



Border Security Forces / SSB Personnel

- Faster identification of potential intrusions
- Reduced dependence on continuous manual monitoring
- Better situational awareness across multiple camera feeds

Administrators / Deployment Teams

- Intelligence layer over existing CCTV infrastructure
- Lower dependency on specialized surveillance hardware
- Scalable deployment across strategic locations

Command & Control Teams

- Centralized incident visibility
- Real-time alerts with supporting evidence
- Faster investigation through searchable event history

SECURITY IMPACT



Earlier Detection

AI continuously analyses video and highlights events requiring attention.

Faster Response

Critical incidents reach operators in real time instead of relying only on manual observation.

Better Situational Awareness

Person, vehicle, movement, zone and event information can be viewed together.

Evidence-Driven Investigation

Alerts can be associated with snapshots, video clips, timestamps and event metadata.



ECONOMIC IMPACT

Lower Infrastructure Cost

Leverages existing CCTV cameras instead of requiring complete smart-camera replacement.

Reduced Hardware Dependency

Advanced analytics are delivered primarily through software and computing infrastructure.

Scalable Investment

Deployment can grow incrementally—from a small camera group to multiple locations.

Operational Efficiency

Operators can focus attention on prioritized incidents instead of continuously watching every feed.



SOCIAL & ENVIRONMENTAL IMPACT

Safer Border Installations

Strengthens surveillance around vulnerable and strategic locations.

Reduced Human Monitoring Burden

AI assists personnel by filtering large volumes of video into relevant events.

Less Hardware Replacement

Reusing existing cameras can reduce unnecessary replacement of functioning surveillance infrastructure.

Remote-Site Support

Edge-based processing can help maintain local intelligence even where connectivity is unreliable.



MORE COVERAGE → LESS MANUAL MONITORING → FASTER DETECTION → FASTER RESPONSE
From Watching Cameras → To Acting on Intelligence

1. Smart India Hackathon — Problem Statement 26187

AI-Based Intelligent Video Analytics Platform for Border Surveillance

<https://www.sih.gov.in/>

Reference: Official SIH Problem Statement

2. Ultralytics YOLO — Object Detection & Tracking

Reference for real-time person, vehicle and object detection.

<https://docs.ultralytics.com/>

3. ByteTrack — Multi-Object Tracking

Research reference for maintaining object identities across video frames.

<https://arxiv.org/abs/2110.06864>

4. ONVIF — IP Video Interoperability

Reference for standardized communication and interoperability with IP-based CCTV systems.

<https://www.onvif.org/profiles/>

5. PaddleOCR — Optical Character Recognition

Reference for OCR-based number-plate text recognition and scene-text processing.

<https://github.com/PaddlePaddle/PaddleOCR>

6. OpenCV — Computer Vision Library

Reference for video processing, frame manipulation and computer-vision operations.

<https://opencv.org/>

7. FastAPI — Backend API Framework

Reference for building high-performance backend APIs and AI-service integration.

<https://fastapi.tiangolo.com/>

8. Docker — Containerized Deployment

Reference for portable and reproducible deployment of application services.

<https://docs.docker.com/>

